

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Step	Action Taken	Observation
1	Shut down the computer and unplugged the power cable.	The system powered off safely.
2	Opened the desktop side panel (or laptop service cover).	The RAM slots were visible on the motherboard.
3	Released the retaining clips on both sides of the RAM module.	The RAM module popped up and was easy to remove.
4	Removed the RAM module and checked for dust or damage.	No visible damage; only a small amount of dust was present.
5	Reinserted the RAM module, aligning the notch correctly, and pressed until the clips locked into place.	The module was seated securely with both clips locked.
6	Closed the case, reconnected the power cable, and turned on the computer.	The computer booted normally and recognized the installed RAM.

Issues Observed

- No physical damage was found on the RAM module or slot.
- The RAM was reseated successfully.
- The computer started normally after reassembly without any POST errors or beep codes.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix. Hint: Think about display issues, driver checks, and physical connections.

Step	Action	Purpose
1. Identify the symptoms	Check for no display, screen flickering, graphical artifacts, crashes, or poor gaming performance.	Helps determine whether the issue is related to the GPU.
2. Check physical connections	Ensure the graphics card is fully seated in the PCIe slot and all PCIe power cables are securely connected. Also check the monitor cable.	Rules out loose connections as the cause.
3. Inspect the GPU	Look for dust buildup, damaged components, and verify that the GPU cooling fans are spinning properly.	Detects overheating or physical damage.
4. Check or reinstall graphics drivers	Update, reinstall, or roll back the GPU drivers if the problem started after a driver update.	Fixes issues caused by corrupted or incompatible drivers.

Step	Action	Purpose
5. Test the GPU	Try the GPU in another compatible PC, or install a known-good GPU in the system.	Confirms whether the graphics card itself is faulty.
6. Confirm the fix	After repairing or replacing the GPU, boot the PC and test games or graphics-intensive applications.	Verifies that the display is stable and the issue has been resolved.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Step	Action	Expected Result
1	Shut down the computer and unplug the power cable.	System is completely powered off.
2	Open the computer case or laptop service panel.	Storage drive (SSD/HDD) is accessible.
3	Disconnect the SATA data cable (and SATA power cable if applicable) or carefully remove the NVMe SSD.	The storage device is disconnected.
4	Turn on the computer and enter the BIOS/UEFI setup.	The BIOS does not detect the SSD/HDD.
5	Try to boot the computer.	The system may display " No Boot Device Found ", " Operating System Not Found ", or " Reboot and Select Proper Boot Device. "
6	Turn off the computer again and unplug the power.	Safe to reconnect the storage device.
7	Reconnect the SATA/NVMe drive securely and ensure all cables are firmly attached.	Storage device is properly connected.
8	Power on the computer and enter the BIOS again.	The BIOS detects the SSD/HDD correctly.
9	Boot into the operating system.	The PC starts normally without storage-related errors.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Scenario: PC Randomly Restarts

Component to Check First: RAM (Memory)

Reason:

Faulty or improperly seated RAM is one of the most common causes of random restarts. If the RAM has errors or a poor connection, the system can become unstable and restart unexpectedly.

Diagnostic Steps for RAM

Step	Action	Purpose
1	Shut down the PC and disconnect the power.	Ensures safety before handling hardware.
2	Remove and reinsert the RAM modules securely.	Checks for loose or improperly seated RAM.
3	Inspect the RAM and slots for dust or damage.	Identifies physical issues that may affect the connection.
4	Test one RAM module at a time in different slots.	Determines whether a RAM stick or motherboard slot is faulty.
5	Boot the PC after each test and observe its behavior.	Checks if the random restart problem is resolved.
6	If available, run a memory diagnostic tool (such as Windows Memory Diagnostic or MemTest86).	Detects memory errors that are not visible through physical inspection.

Conclusion

If the PC no longer restarts after reseating or replacing the RAM, the issue was likely caused by a faulty or loose memory module. If the problem continues, the next components to check are the **power supply (PSU)**, **GPU**, and **storage drive**.